

Rachel Lin

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EDUCATION/AWARDS

Amherst College | Amherst, MA

May 2025

- Bachelor of Arts: Computer Science
- Relevant Coursework: Computing and the Arts, Statistics via Modeling, Linear Algebra via Applications, Data Structures, Algorithms, Artificial Intelligence, Data Science, Machine Learning, Data Mining

Major League Hacking CivicHacks Hackathon | *Third Place, Best Accessibility Hack*

January 2023

EXPERIENCE

Keelung For a Walk Program | Keelung, Taiwan

February 2026 - Present

Frontend Developer (Volunteer)

- Developed gameplay features in a *Vue.js SPA*, including timer tracking, configurable completion screens, and WebAR challenges (*AR.js*) supporting 200+ challenge codes per hunt.
- Migrated legacy URLs/assets and integrated Firebase configuration and client state persistence.
- Configured GitHub Actions CI/CD for automated builds and deployment to Firebase Hosting.

Full Stack Developer @ IHAI Time (Initiative) | Remote

July 2025 - Present

- Implemented cross-functional workflow automation and data dashboards for MVP timebanking system, improving project turnaround time by ~12%
- Collaborated with product, engineering, and outreach teams to establish partnerships with 5+ local nonprofits and student organizations, expanding platform reach and pilot adoption

Data Mining Intern | Amherst, MA

January 2025 - May 2025

- Scraped and processed ~10,000 tweets using *Selenium* and applied NLP techniques (logistic regression, random forest, KDE) to perform large-scale sentiment analysis on real-time metadata of feminist hashtags across Latin America
- Built 15+ interactive visualizations (time series, sentiment trends, engagement metrics) and delivered several data-driven reports highlighting key user behavior patterns and hashtag performance to discover feminist trends

COSC-112 Teaching Assistant | Amherst, MA

September 2023 - December 2023

- Manage afternoon labs and hold evening TA sessions weekly for ~40 total students in Intro to Computer Science II to provide assistance with assignments, projects, and key concepts. Held and managed 12+ TA labs and sessions for a total of over 40 hours.

Research Fellowship → Gregory Call Academic Intern | Amherst, MA

June 2023 - December 2023

- Conduct research on Queuing theory models and simulate pass-and-swap queues using python and mathematically analyze queuing policies to better predict system performance by 22%
- Simulated a Jackson network of Starbucks customer queues (M/M/k) using Python, analyzing customer behavior (joining vs. balking) in a partially observable system and improving queue throughput by 17%.

PROJECTS

Anti-Brainrot | *Electron (IPC), React/Vite, JS, TensorFlow.js (COCO-SSD), Node.js* | [Link](#)

April 2026

- Built a desktop productivity app for real-time phone detection from the user's webcam using computer vision: implemented ~100ms detection intervals, parallel crop-region inference, and hysteresis/cooldown logic to reduce false positives.
- Designed multi-window Electron architecture with IPC communication: automatic meme/GIF + audio interruptions when phone usage is detected, with local-only processing for privacy-safe focus sessions and session tracking.

Photobooth+ | *React, Vite*

November 2025

- Developed a React-based photobooth app with real-time image processing and automated rendering pipeline, improving user interaction flow and reducing capture latency
- Implemented real-time photo capture and print-style UI effects with modular CSS and responsive grid layouts.

Being Human in STEM Database Project | *PostgreSQL, Django, React.js, Node.js*

September 2023

- Created database management system for HSTEM student-led initiative on campus that stored 400+ student projects; web architecture developed via REST API with Django REST framework to manage backend

OTHER/ACTIVITIES

The Remedy Project (Civic Advocacy Organization) | Remote

January 2023 - February 2024

Lead Data & IT Team Member (Volunteer)

- Managed backend operations of the action-page platform using NationBuilder, supporting supporter data workflows, CRM segmentation, donation pipelines, and campaign automation for 1,000+ community members
- Collaborated with co-founders to improve website architecture and user experience by customizing templates with Jinja, HTML, CSS, and JavaScript, building dynamic landing pages and streamlining action-page conversions

SKILLS & INTERESTS

LANGUAGES (Technical): Python, Javascript, Java, HTML/CSS, JS, SQL (Postgres), Typescript, R

FRAMEWORKS: React.js, Node.js, Vue.js

SKILLS & TOOLS: Github/Git, VSCode, RStudio, Tableau, Vercel, Xcode, AWS (S3), Excel